

Delegation in Strategic Environments and Equilibrium Uniqueness

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Motivation

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- new channel: **algorithmic delegation**
 - ▶ market participants authorize an algorithmic intermediary to act on their behalf
 - ▶ algorithms can coordinate actions at scale
- regulators are increasingly concerned this may produce unintended outcomes

Case study: RealPage antitrust litigation

- **RealPage**: revenue-management software used by major US landlords
- sued by the US Department of Justice in 2024
- core allegation: facilitated coordination on higher rents among rival landlords

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⇒ algorithmic rent-setting now banned in NY, Colorado, California, and many US cities

Broader concerns

- non-competitive outcomes may arise even if each market participant uses their own algorithm

“Absent concerted action, independent adoption of the same or similar pricing algorithms is unlikely to lead to antitrust liability even if it makes interdependent pricing more likely.”

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- yet the scope of the problem remains poorly understood

“This body of research is in its infancy.”

FTC Staff, *The Rise of Surveillance Pricing* (Issue Spotlight, Jan. 2025)

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Our Question

when does a strategic interaction deliver a **unique equilibrium outcome** even if players can [delegate](#) play to an intermediary with substantial coordinating power?

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- two extremes of intermediary power:
 - ▶ **weak**: cannot punish deviators $\Rightarrow$ **coarse correlated equilibrium** (CCE)
 - ▶ **strong**: can coordinate adversarial punishments $\Rightarrow$ **individually rational correlated profiles** (IRCP)

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Main contribution: characterizations of games with unique CCE or unique IRCP

- $\Rightarrow$ tractable sufficient conditions for uniqueness NE and CE

Related literature

- **Correlated solution concepts:** CE (Aumann, 1974, 1987); CCE (Hannan, 1957; Moulin and Vial, 1978); IRCP (Myerson, 1991, Chapter 6)
- **Foundations:** pre-play communication $\rightarrow$ CE (Bárány, 1992; Ben-Porath, 1998; Gerardi, 2004; Lehrer and Sorin, 1997); learning: external regret $\rightarrow$ CCE (Hannan, 1957), internal regret $\rightarrow$ CE (Foster and Vohra, 1997; Fudenberg and Levine, 1999; Hart and Mas-Colell, 2000); folk theorem $\rightarrow$ IRCP (Benoit, Krishna, et al., 1984; Fudenberg and Maskin, 1986; Aumann and Shapley, 1976);
- **Unique CE:** robust to payoff perturbations (Kajii and Morris, 1997; Einy, Haimanko, and Lagziel, 2022); convex potential (Neyman, 1997; Ui, 2008; Cao, Tan, and Zhou, 2025); zero-sum (Forges, 1990; Viossat, 2006; Kattwinkel, Niemeyer, Preusser, and Winter, 2022; Kartik, Squintani, and Tinn, 2024); auctions (Feldman, Lucier, and Nisan, 2016)
- **Unique CCE:** strict social concavity (Even-Dar, Mansour, and Nadav, 2009; Hart and Mas-Colell, 2015); non-atomic congestion (Koessler et al., 2025); other examples (Nadav and Piliouras, 2010; Baudin, Scarsini, and Venel, 2023; Ahunbay and Bichler, 2025)

Roadmap

- ① Model & solution concepts: NE, CE, CCE, IRCP
- ② Unique IRCP
- ③ Unique pure CCE
- ④ Application: contests and potential games
- ⑤ Unique mixed CCE
- ⑥ Extensions and open problems

Model & solution concepts

Delegation as a correlated device

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Key modeling choice: what can the intermediary do **after** an opt-out?

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Two extremes

weak intermediary: cannot punish vs **strong intermediary:** can coordinate adversarial punishments

CCE : weak intermediary

Coarse Correlated Equilibrium (CCE) ([Hannan, 1957](#); [Moulin and Vial, 1978](#))

a distribution $\mu \in \Delta(A)$ is a CCE if for every player i and fixed deviation $a'_i \in A_i$

$$\underbrace{\mathbb{E}_{\mu}[u_i(a)]}_{\text{utility of delegation}} \geq \underbrace{\mathbb{E}_{\mu}[u_i(a'_i, a_{-i})]}_{\text{utility of a fixed action } a'_i}$$

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contrast with Correlated Equilibrium (CE) ([Aumann, 1987](#)):

- CE captures recommendations rather than delegation
- a player can reconsider a recommendation a_i **after** seeing it
- μ is a CE iff for all i and maps $\sigma_i: A_i \rightarrow A_i$

$$\mathbb{E}_{\mu}[u_i(a)] \geq \mathbb{E}_{\mu}[u_i(\sigma_i(a_i), a_{-i})]$$

IRCP: strong intermediary

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- chain of inclusions:

$$\text{NE} \subseteq \text{CE} \subseteq \text{CCE} \subseteq \text{IRCP}$$

Unique IRCP

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Conclusion: in our search for games with a unique IRCP, we can focus on those with pure strict NE

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In fact, dominance is orthogonal to IRCP uniqueness

Example: parking enforcement

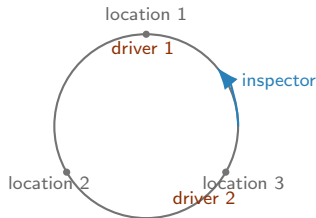
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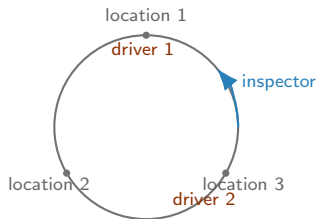
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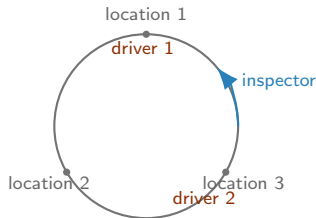


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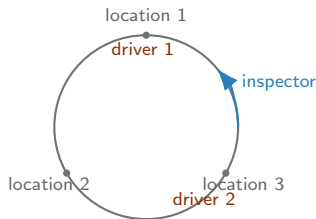


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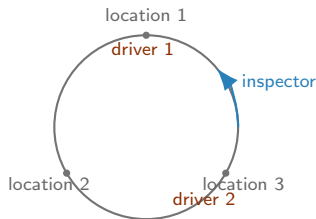


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- **Intuition:** at least one violator will be caught with probability $\geq \frac{1}{2} \Rightarrow$ prefers legal parking

Enforcement games

Enforcement game

A game $H = (A, v)$ is an **enforcement game** if there exists $a^* \in A$ such that

$$v_i(a^*) = 0, \quad v_i(a_i^*, a_{-i}) \geq 0 \quad \forall i, a_{-i}, \quad \sum_i v_i(a) < 0 \quad \forall a \neq a^*.$$

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Lemma

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Proof:

- if $\mu(A \setminus \{a^*\}) > 0$, then expected welfare < 0
- $\Rightarrow$ some player violates IR constraint



Characterization of games with unique IRCP

Theorem 1

A game G has a unique IRCP $\iff G$ is **affinely equivalent** to an enforcement game

- $G = (A, u)$ and $H = (A, v)$ are affine equivalent if $v_i = \gamma_i(u_i + \beta_i)$ with $\gamma_i > 0$ and $\beta_i \in \mathbb{R}$

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the dual to this LP gives γ_i and β_i

□

Unique CCE

Weak delegation: invariances matter

- CCE is invariant to payoff shifts that depend only on **others'** actions
 - ▶ $\Leftarrow$ CCE compares utility differences across own actions

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We first focus on pure CCE and discuss mixed CCE later

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- furthermore, a^* is the unique CCE of G iff a^* is the unique IRCP of the transformed game

$$v_i(a) = u_i(a) - u_i(a_i^*, a_{-i}),$$

that is, one can always take $\beta_i(a_{-i}) = -u_i(a_i^*, a_{-i})$ in strategic equivalence

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- $\Rightarrow$ an easy-to check criterion for unique CCE

Testing for unique pure CCE

Criterion for uniqueness

a^* is a unique CCE iff there exist weights $\gamma_i > 0$ such that for all $a \neq a^*$,

$$\Phi_{a^*}(a) = \sum_i \gamma_i \left(u_i(a) - u_i(a_i^*, a_{-i}) \right) < 0$$

For symmetric games, can take $\gamma_i \equiv 1$

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- uniqueness of CCE additionally requires **multilateral deviations** to be negative-sum in Φ

A simple sufficient condition for unique CCE

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Idea: for strictly concave functions, coordinate-wise maximum $\Rightarrow$ global maximum

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- A_i is compact and convex
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- this class includes linear Cournot and Tullock contests with strictly convex costs
- but there are economically relevant examples beyond

Application: contests & potential games

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Remark: separable Φ is another class where coordinate-wise maximum $\Rightarrow$ global maximum

Ratio-based contests

- symmetric ratio-based contests: $p_1(a) = f\left(\frac{a_1}{a_2}\right)$ and $f(t) + f(1/t) = 1$
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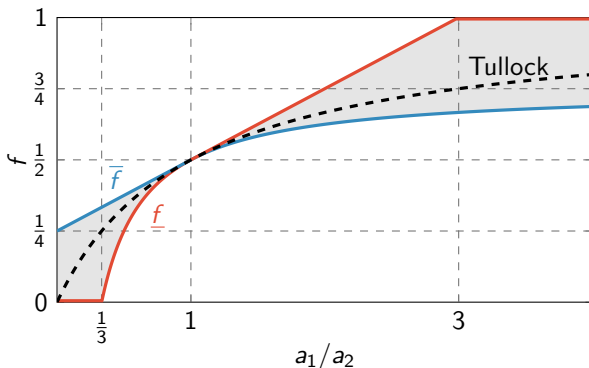
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- a contest is in this class $\Leftrightarrow f$ lies between $\bar{f}$ and $\underline{f}$



Potential games

- a **potential game** is one strategically equivalent to $u_i \equiv P$ (Monderer and Shapley, 1996)
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- separates environments with unique CE from those with unique CCE

Mixed unique CCE

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matching pennies		
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H	$(1, -1)$	$(-1, 1)$
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Conclusion: in contrast to IRCP, unique CCE can be mixed

Unique CCE is “almost pure”

Proposition

If $\text{CCE}(G)$ is unique, then either

- it is a **pure** NE, or
- exactly two players mix over two actions each, and the induced 2×2 game is of **matching-pennies type**.

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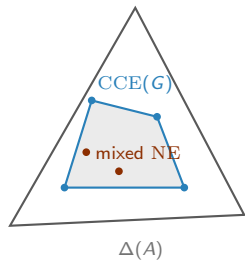
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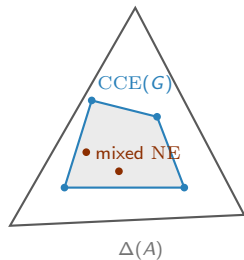
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 - mixed uniqueness is essentially a two-player artifact

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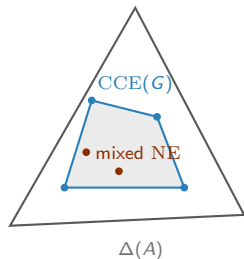
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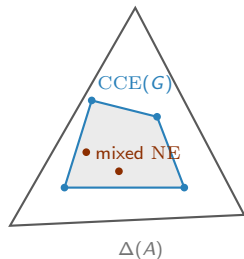
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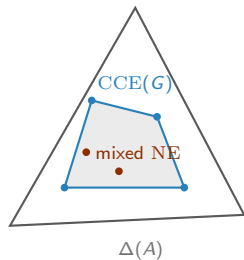
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A quasi-strict NE is extreme in CCE $\iff$ it is pure or 2×2 mixing

Extensions

When does delegation just select among Nash?

- a more permissive benchmark for when delegation doesn't expand outcomes:

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- the intermediary only selects which NE to play

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- quasi-strictness + extremality $\Rightarrow$ pure (2×2 mixing is ruled out for IRCP)
- For generic G , k strict pure NE $\Rightarrow \exists k - 1$ mixed NE (Gül, Pearce, and Stacchetti, 1993)
- A mixed NE requires *indifferences* on its support, which cannot be generated as a convex combination of strict pure NE

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Open question

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“No” in 2×2 games (Calvó-Armengol, 2006) and, more generally, two-person games (Cripps, 1995)

Take-away

- unique delegation outcomes in broad classes of games
- **strong intermediary (IRCP):** uniqueness $\Leftrightarrow$ affine enforcement up to affine equivalence
- **weak intermediary (CCE):** uniqueness $\Leftrightarrow$ enforcement up to strategic equivalence
- **design takeaway:** collusion-proofness is a tractable and permissive constraint
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Thank you!

References I

- Mete Şeref Ahunbay and Martin Bichler. On the uniqueness of bayesian coarse correlated equilibria in standard first-price and all-pay auctions. In *Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 2491–2537. SIAM, 2025.
- Robert J Aumann. Subjectivity and correlation in randomized strategies. *Journal of mathematical Economics*, 1(1):67–96, 1974.
- Robert J Aumann. Correlated equilibrium as an expression of bayesian rationality. *Econometrica: Journal of the Econometric Society*, pages 1–18, 1987.
- Robert J Aumann and Lloyd S Shapley. Long-term competition—a game-theoretic analysis. In *reprinted in Essays in Game Theory: In Honor of Michael Maschler (ed. N. Megiddo), Springer, 1994, pp. 1–15*. 1976.
- Imre Bárány. Fair distribution protocols or how the players replace fortune. *Mathematics of Operations Research*, 17(2):327–340, 1992.
- Lucas Baudin, Marco Scarsini, and Xavier Venel. Strategic behavior and no-regret learning in queueing systems. *arXiv preprint arXiv:2302.03614*, 2023.

References II

- Elchanan Ben-Porath. Correlation without mediation: Expanding the set of equilibrium outcomes by “cheap” pre-play procedures. *Journal of Economic Theory*, 80(1):108–122, 1998.
- Jean-Pierre Benoit, Vijay Krishna, et al. Finitely repeated games. 1984.
- Antoni Calvó-Armengol. The set of correlated equilibria of 2x2 games. *mimeo*, 2006.
- Zhigang Cao, Zhibin Tan, and Jinchuan Zhou. Correlated equilibrium of games with concave potential functions. *Operations Research Letters*, 60:107241, 2025.
- M Cripps. Extreme correlated and Nash equilibria in two-person games, 1995.
- Ezra Einy, Ori Haimanko, and David Lagziel. Strong robustness to incomplete information and the uniqueness of a correlated equilibrium. *Economic Theory*, 73(1):91–119, 2022.
- Eyal Even-Dar, Yishay Mansour, and Uri Nadav. On the convergence of regret minimization dynamics in concave games. In *Proceedings of the forty-first annual ACM symposium on Theory of computing*, pages 523–532, 2009.
- Michal Feldman, Brendan Lucier, and Noam Nisan. Correlated and coarse equilibria of single-item auctions. In *Web and Internet Economics: 12th International Conference, WINE 2016, Montreal, Canada, December 11-14, 2016, Proceedings 12*, pages 131–144. Springer, 2016.

References III

- Françoise Forges. Correlated equilibrium in two-person zero-sum games. *Econometrica* (1986-1998), 58 (2):515, 1990.
- Dean P Foster and Rakesh V Vohra. Calibrated learning and correlated equilibrium. *Games and Economic Behavior*, 21(1-2):40–55, 1997.
- Drew Fudenberg and David K Levine. Conditional universal consistency. *Games and Economic Behavior*, 29(1-2):104–130, 1999.
- Drew Fudenberg and Eric Maskin. The folk theorem in repeated games with discounting or with incomplete information. *Econometrica*, 54(3):533–554, 1986.
- Dino Gerardi. Unmediated communication in games with complete and incomplete information. *Journal of Economic Theory*, 114(1):104–131, 2004.
- Faruk Gül, David Pearce, and Ennio Stacchetti. A bound on the proportion of pure strategy equilibria in generic games. *Mathematics of Operations Research*, 18(3):548–552, 1993.
- James Hannan. Approximation to bayes risk in repeated play. *Contributions to the Theory of Games*, 3 (2):97–139, 1957.

References IV

- John C Harsanyi. Oddness of the number of equilibrium points: a new proof. *International Journal of Game Theory*, 2:235–250, 1973.
- Sergiu Hart and Andreu Mas-Colell. A simple adaptive procedure leading to correlated equilibrium. *Econometrica*, 68(5):1127–1150, 2000.
- Sergiu Hart and Andreu Mas-Colell. Markets, correlation, and regret-matching. *Games and Economic Behavior*, 93:42–58, 2015.
- Atsushi Kajii and Stephen Morris. The robustness of equilibria to incomplete information. *Econometrica*, 65(6):1283–1309, 1997. ISSN 00129682, 14680262. URL <http://www.jstor.org/stable/2171737>.
- Navin Kartik, Francesco Squintani, and Katrin Tinn. Information revelation in constant-sum games: Elections and beyond. *arXiv preprint arXiv:2406.17084*, 2024.
- Deniz Kattwinkel, Axel Niemeyer, Justus Preusser, and Alexander Winter. Mechanisms without transfers for fully biased agents. In *Proceedings of the 23rd ACM Conference on Economics and Computation*, pages 332–333, 2022.

References V

- Frédéric Koessler, Marco Scarsini, and Tristan Tomala. Correlated equilibria in large anonymous bayesian games. *Mathematics of Operations Research*, 50(3):2157–2174, 2025.
- Ehud Lehrer and Sylvain Sorin. One-shot public mediated talk. *Games and Economic Behavior*, 20(2): 131–148, 1997.
- Dov Monderer and Lloyd S Shapley. Potential games. *Games and economic behavior*, 14(1):124–143, 1996.
- Hervé Moulin and J P Vial. Strategically zero-sum games: the class of games whose completely mixed equilibria cannot be improved upon. *International Journal of Game Theory*, 7:201–221, 1978.
- Roger B Myerson. *Game Theory: Analysis of Conflict*. Harvard University Press, 1991.
- Uri Nadav and Georgios Piliouras. No regret learning in oligopolies: Cournot vs. Bertrand. In *International Symposium on Algorithmic Game Theory*, pages 300–311. Springer, 2010.
- Abraham Neyman. Correlated equilibrium and potential games. *International Journal of Game Theory*, 26:223–227, 1997.
- Takashi Ui. Correlated equilibrium and concave games. *International Journal of Game Theory*, 37:1–13, 2008.

References VI

Yannick Viossat. The geometry of Nash equilibria and correlated equilibria and a generalization of zero-sum games. Technical report, SSE/EFI Working Paper Series in Economics and Finance, 2006.